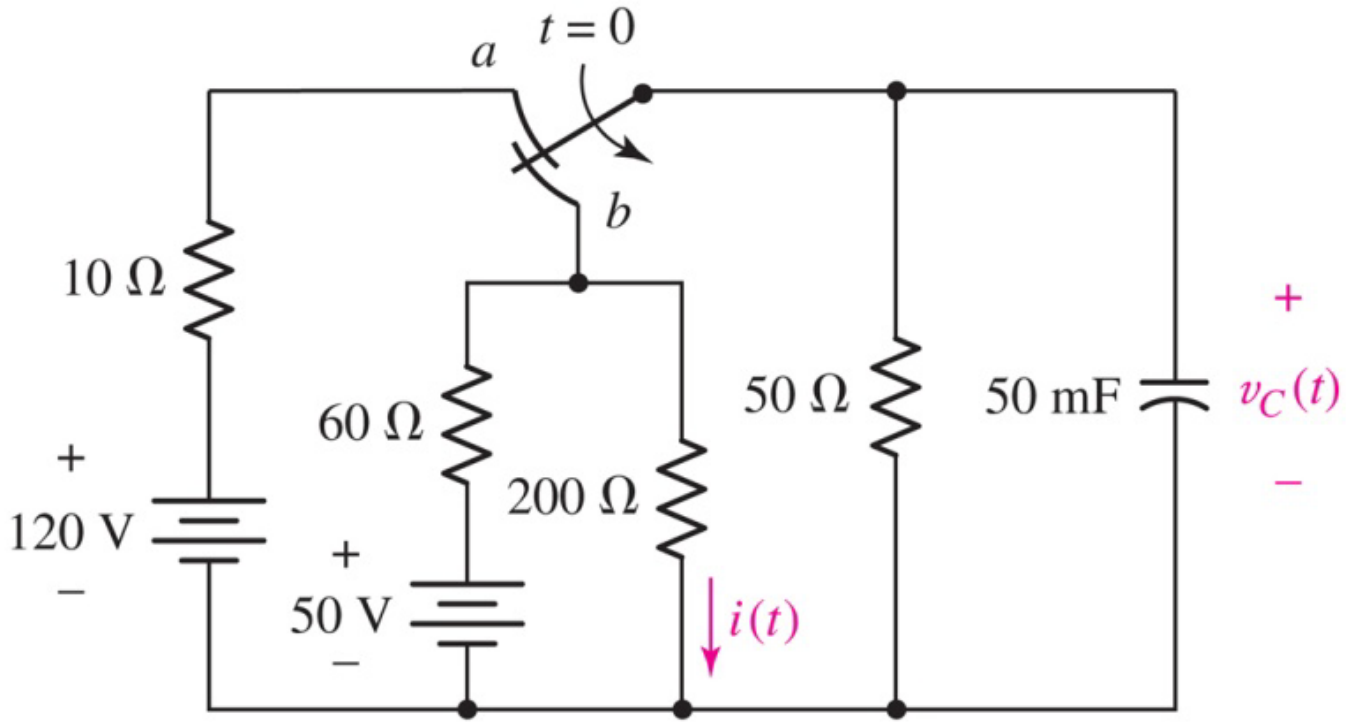
Tutorial Question 1 (Example 4 continued)

Find the current $i(t)$ through the 200Ω resistor all the time.



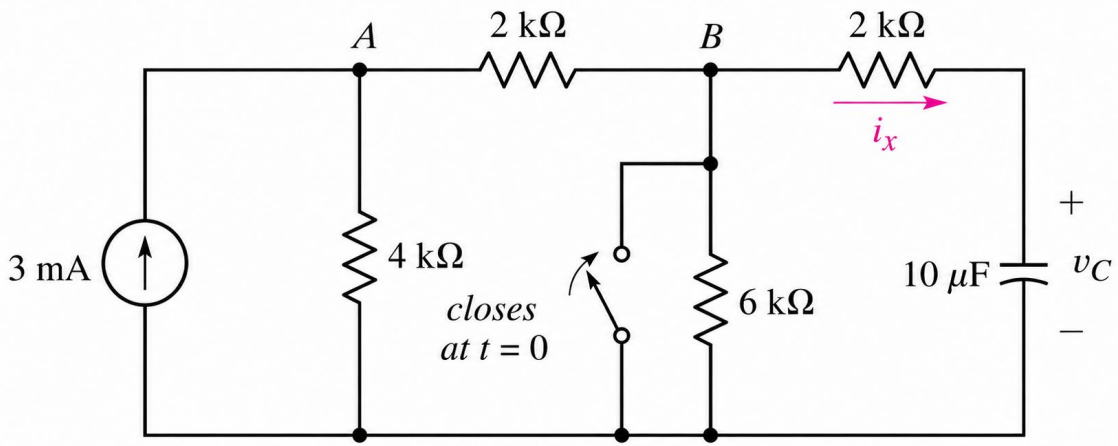
Tutorial question 2

A voltage source, $v_s = 20u(t)$ V, is in series with a 200 Ω resistor and a 4 H inductor. Find the magnitude of the inductor current at t equal to (a) 0^-; (b) 0^+; (c) 8 ms; (d)15 ms.



Tutorial Question 3

The switch has been open for a very long time and closes at t = 0. The capacitor voltage and current are defined as shown.



(a) Determine $v_C(0^-)$, $v_C(0^+)$ and $v_C(\infty)$.
(b) Find the time constant and $v_C(t)$ for $t \geq 0$.
(c) Find $i_x(0^+)$ and state the actual current direction.
(d) When does the capacitor voltage first reach 2 V?
(e) Find the total energy dissipated in the right-hand 2 kΩ resistor during the discharge.

Solution to tutorial question 1(Example 4 continued)

Find current $i(t)$ in the 200Ω resistor all the time.

Step 1: Calculate the time constant.

$$\tau_{eq} = R_{eq}C = 1.2s$$

Step 2: Determine the initial condition $i(0^+)$.

$$v_c(0^+) = v_c(0^-) = 100\text{V} \qquad i(0^+) = \frac{v_c(0^+)}{200} = 0.5\text{A}$$

Step 3: Determine the final condition $i(\infty)$.

$$i(\infty) = \frac{v_c(\infty)}{200} = \frac{20}{200} = 0.1\text{A}$$

Step 4: Final response. $i = i(\infty) + [i(0^+) - i(\infty)]e^{-t/\tau}$

For $t \geq 0$

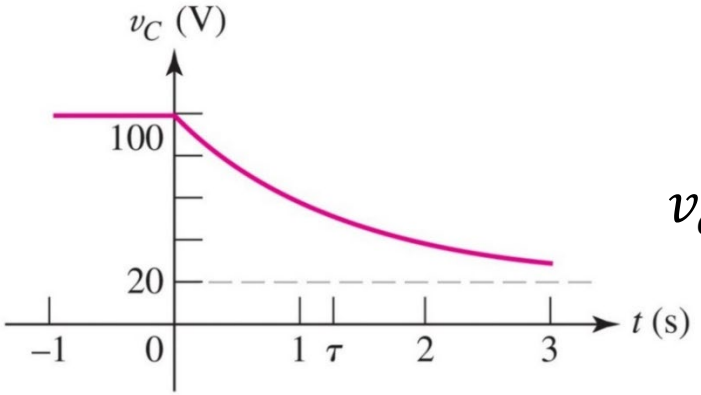
$$i(t) = 0.1 + [0.5 - 0.1]e^{-t/1.2}$$

For $t < 0$

$$i(t) = \frac{50}{260} = 0.1923\text{A}$$

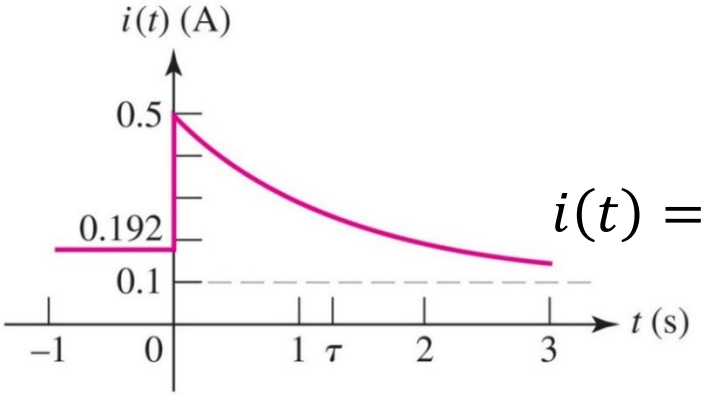


Solution to tutorial question 1 (Example 4 continued)



(a)

$$v_c(t) = 100u(-t) + (20 + 80e^{-\frac{t}{1.2}})u(t)\text{V}$$



(b)

$$i(t) = 0.192u(-t) + (0.1 + 0.4\,e^{-t/1.2})u(t)\text{A}$$

- According to the above two figures, we can see that the capacitor voltage cannot jump and the resistor current (capacitor current) can jump.
- You can also find the 200Ω resistor current directly by $i(t) = \frac{v_c(t)}{200}$ for $t \geq 0$.

Solution to Tutorial question 2

(a) At t= 0^-, $i_L(0^-) = \frac{0\ \text{V}}{200\Omega} = 0\ \text{A}$

(b) At t= 0^+, $i_L(0^+) = i_L(0^-) = 0\ \text{A}$

(c) $i_L(t) = \frac{V_0}{R}(1 - e^{-\frac{tR}{L}}) = 0.1\left(1 - e^{-\frac{t}{0.02}}\right)$

$i_L(t = 0.008) = 0.1\left(1 - e^{-\frac{0.008}{0.02}}\right) = 0.03297\ \text{A}$

(d) $i_L(t = 0.015) = 0.1\left(1 - e^{-\frac{0.015}{0.02}}\right) = 0.05276\ \text{A}$



Solution to Tutorial Question 3

(a) Before switching, the capacitor is an open circuit at steady state.

$$v_C(0^-) = (3\text{ mA})\frac{4}{4+2+6}(6\text{ k}\Omega) = 6\text{ V}$$

$$v_C(0^+) = 6\text{ V}, \qquad v_C(\infty) = 0\text{ V}$$

(b) After switching, only the right-hand 2 kΩ resistor sets τ.

$$\tau = RC = (2\text{ k}\Omega)(10\text{ μF}) = 20\text{ ms}$$

$$v_C(t) = 6e^{-\frac{t}{20\text{ ms}}}\text{ V}, \qquad t \geq 0$$

(c) Initial current:

$$i_x(0^+) = \frac{0-6}{2\text{ k}\Omega} = -3\text{ mA}$$

The actual current flows from right to left.

(d) Time to reach 2 V:

$$2 = 6e^{-\frac{t}{20\text{ ms}}} \quad \Rightarrow \quad t = 20\ln 3 = 21.97\text{ ms}$$

(e) All the initial capacitor energy dissipates in this resistor.

$$W_R = \frac{1}{2}Cv_C{}^2(0^+) = \frac{1}{2}(10\text{ μF})(6^2) = 180\text{ μJ}$$

